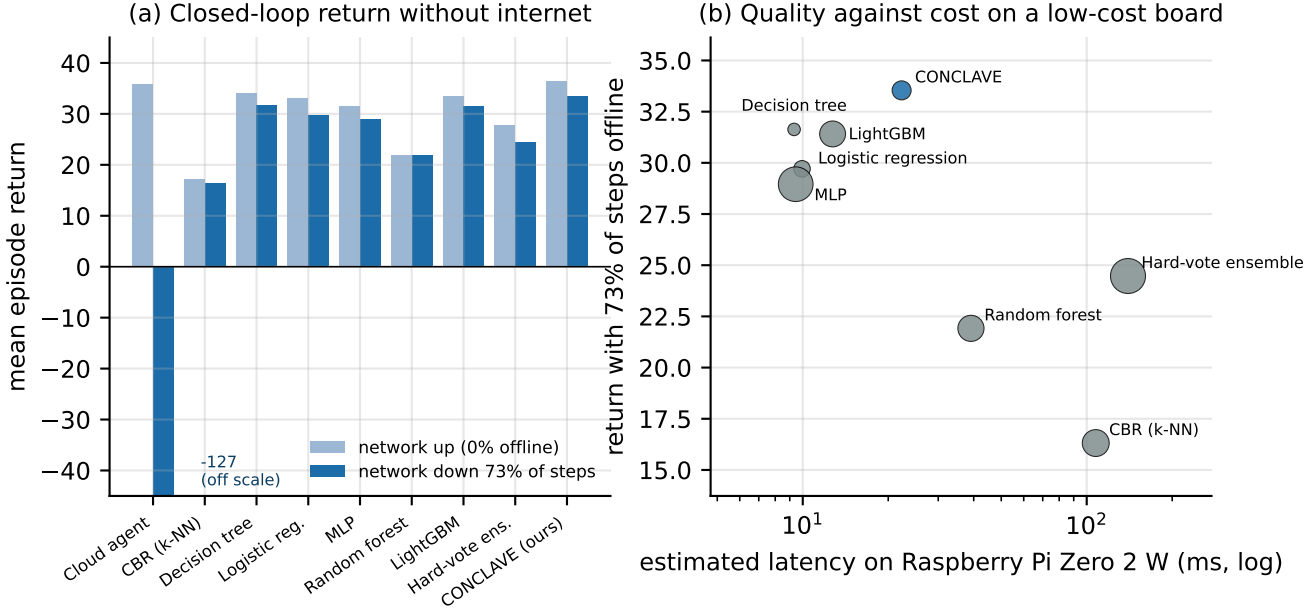
(a) Closed-loop return without internet
mean episode return
40
30
20
10
0
−10
−20
−30
−40
-127
(off scale)
network up (0% offline)
network down 73% of steps
Cloud agent
CBR (k-NN)
Decision tree
Logistic reg.
MLP
Random forest
LightGBM
Hard-vote ens.
CONCLAVE (ours)
(b) Quality against cost on a low-cost board
return with 73% of steps offline
35.0
32.5
30.0
27.5
25.0
22.5
20.0
17.5
15.0
CONCLAVE
Decision tree
LightGBM
Logistic regression
MLP
Hard-vote ensemble
Random forest
CBR (k-NN)
10^1
10^2
estimated latency on Raspberry Pi Zero 2 W (ms, log)